Applied Calculus IV

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Reference & Copyright Notice

Week 10

Review: Chapter 1 to 4

Contents

Midterm Exam Notification

Dear Students,

This is a reminder that the midterm exam will take place on November 30.

Please note the exam scope clearly:

> The exam covers Chapters 1, 2, and 4 of the textbook. For Chapter 4, only the subsections we covered in class—4.1.1, 4.2, 4.3, 4.4, and 4.6—will be included.
>
> Additionally, all content in the lecture slides from week 1 to week 10, including exercises and examples, is also part of the exam, so please review these materials carefully.

We recommend you organize your study plan in advance to cover both the textbook chapters and slides. If you need any help, you are very welcome to my office (Mon & Wed : 14:00 to 18:00)

Best Regards

Exercises

Q. Find the solution of the following differential equations

1. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = y + \sin x$

2. $\dfrac{\mathrm{d}x}{\mathrm{d}t} + 3x = \mathrm{e}^{2t}$

3. $\dfrac{\mathrm{d}s}{\mathrm{d}t} = -s\cos t + \dfrac{1}{2}\sin 2t$

4. $\dfrac{\mathrm{d}y}{\mathrm{d}x} - \dfrac{n}{x}y = \mathrm{e}^{x}x^{n}$

5. $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{1-2x}{x^2}y - 1 = 0$

6. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^4+y^3}{xy^2}$

7. $\dfrac{\mathrm{d}y}{\mathrm{d}x} - \dfrac{2y}{x+1} = (x+1)^3$

8. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y}{x+y^3}$

9. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{ay}{x} + \dfrac{x+1}{x}$

10. $x\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = x^3$

11. $\dfrac{\mathrm{d}y}{\mathrm{d}x} + xy = x^3y^3$

12. $(y\ln x - 2)y\mathrm{d}x = x\mathrm{d}y$

13. $2xy\mathrm{d}y = (2y^2 - x)\mathrm{d}x$

14. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{e}^y + 3x}{x^2}$

15. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{xy + x^3y^3}$

Hint :

1,2,3,4,5,7,9,10 directly can be matched with this form

$$\boxed{\frac{dy}{dx} + P(x)y = f(x),}$$

6, 11, 13 are Bernoulli's equation

14 let $u = e^y$ and it will be in the form of Bernoulli's equation

8, 15 rearrange them, somehow y get the role of the independent variable

Q. Find the solution of the following integral equations

$$y = \mathrm{e}^{x} + \int_0^x y(t)\mathrm{d}t$$

Solution:

Step 1. Find the DE

$$y = \mathrm{e}^x + \int_0^x y(t)\mathrm{d}t \xrightarrow{\text{Find the derivative of both side}} y' = e^x + y$$

Step2. Solve the DE

Q: Suppose the function $\varphi(t)$ is continuous on $-\infty < t < +\infty$, $\varphi'(0)$ exists and satisfies the relation

$$\varphi(t+s) = \varphi(t)\varphi(s)$$

Find the function $\varphi(t)$.

Solution:

Step 1: Determine $\varphi(0)$

Substitute $t = s = 0$: $\varphi(0) = \varphi(0)^2 \rightarrow \varphi(0)\big(1 - \varphi(0)\big) = 0 \rightarrow \varphi(0) = 0\,,1$

Step 2: Use definition of derivative to derive the differential equation

$$\varphi'(t) = \lim_{h\to 0} \frac{\varphi(t+h)-\varphi(t)}{h} = \lim_{h\to 0} \frac{\varphi(t)\varphi(h)-\varphi(t)}{h} = \varphi(t) \underbrace{\lim_{h\to 0} \frac{\varphi(h)-1}{h}}_{\varphi'(0)}$$

$a = \varphi'(0)$

because $\varphi'(0) = \lim_{h\to 0} \dfrac{\varphi(h) - \cancel{\varphi(0)}^{\,=1}}{h}$

$$\varphi'(t) = a\varphi(t)$$

we obtained a differential equation

Step 3: Solve the differential equation and apply the initial condition

$$\xrightarrow{\text{By integration}} \varphi(t) = c_1 e^{at} \xrightarrow{\varphi(0) = 1} \varphi(t) = e^{at} \longrightarrow \varphi(t) = e^{\varphi'(0)\cdot t}$$

What if $\varphi(0) = 0$:

$\xrightarrow{\text{using the given formula}}$ If $s = 0 \quad \rightarrow \quad \varphi(t+0) = \varphi(t)\varphi(0) = 0 \quad \rightarrow \quad \varphi(t) = 0$ it gives the trivial solution

Differential equation of capacitor voltage in charging process (for an RC circuit: resistor R and capacitor C in series with a DC voltage source E)

$$\boxed{RC\frac{du_C}{dt} + u_C = E}$$

Differential equation of current (for an RL circuit: resistor R and inductor L in series with a DC voltage source E)

$$\boxed{L\frac{\mathrm{d}I}{\mathrm{d}t} + RI = E}$$

$$\frac{dy}{dx} + P(x)y = f(x)$$

Q. For the RL circuit shown in the Figure.

a. Find the current (I) on the inductor L 10 seconds after switch S_1 is closed;

b. After 10 seconds, close S_2, and find the current (I) on the inductor L after 20 seconds.

Hint:

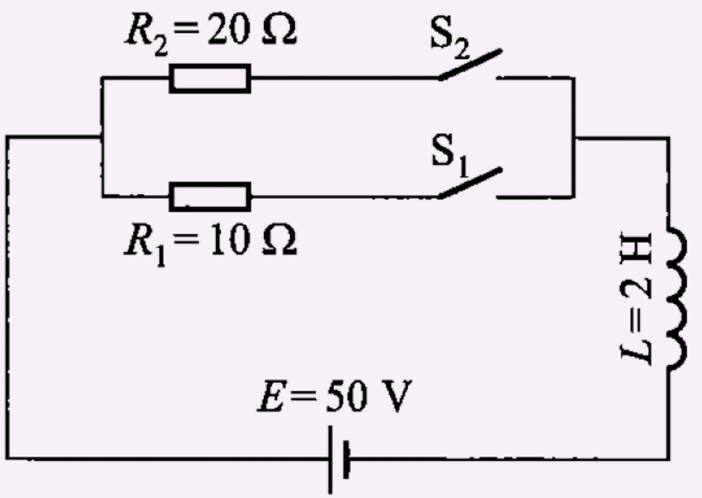
$$\boxed{L\frac{dI}{dt} + RI = E}$$

This DE describes how the inductor current changes over time during charging/discharging.

L: Inductance (Henries, H),

R: Resistance (ohms, Ω),

E: Source voltage (volts, V).



the RL circuit

Solution part a:

$$\boxed{L\frac{\mathrm{d}I}{\mathrm{d}t} + RI = E}$$

Step 1. Find DE when $L = 2, E = 50, R = 10$:

$$2\frac{\mathrm{d}I}{\mathrm{d}t} + 10I = 50 \implies \boxed{\frac{\mathrm{d}I}{\mathrm{d}t} + 5I = 25}$$

Step 2. Find the general solution

$$\mu(t) = e^{\int P(t)\,\mathrm{d}t} = e^{\int 5\,\mathrm{d}t} = e^{5t}$$

$$e^{5t}\frac{\mathrm{d}I}{\mathrm{d}t} + 5e^{5t}I = 25e^{5t}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(Ie^{5t}\right) = 25e^{5t} \rightarrow \boxed{I(t) = 5 + ce^{-5t}}$$

Step 3. Use the initial condition $I(0) = 0$ (initial current is 0)

$$0 = 5 + c \implies c = -5$$

$$\rightarrow \boxed{I(t) = 5(1 - e^{-5t})}$$

Step 4. Find current at t =10 s

$$\rightarrow I(10) = 5(1 - e^{-50}) \approx 5\ \mathrm{A}$$

Solution part b:

$$\boxed{L\frac{\mathrm{d}I}{\mathrm{d}t} + RI = E}$$

Step 1. Find DE when $L = 2, E = 50, R = 20$: $2\frac{\mathrm{d}I}{\mathrm{d}\tau} + 20I = 50 \implies \boxed{\frac{\mathrm{d}I}{\mathrm{d}\tau} + 10I = 25}$

Step 2. Find the general solution

$$\mu(t) = e^{\int P(t)\,\mathrm{d}t} = e^{\int 10\,\mathrm{d}t} = e^{10t}$$

$$e^{10t}\frac{\mathrm{d}I}{\mathrm{d}t} + 10e^{10t}I = 25e^{10t}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(Ie^{10t}\right) = 25e^{10t} \rightarrow \boxed{I(t) = \frac{5}{2} + ce^{-10t}}$$

Step 3. Use the initial condition $I(0) = 5$ (initial current is 5) $\rightarrow 5 = 2.5 + ce^{-100} \implies c = 2.5e^{100}$

$$\rightarrow I(t) = 2.5 + 2.5e^{100}\cdot e^{-10t} = \boxed{2.5 + 2.5e^{-10(t-10)}}$$

Step 4. Find current at t =10 + 20 s,

(however we should replace t =20 s in formula) $\rightarrow I(30) = 2.5 + 2.5e^{-10(20)} = 2.5(1 + e^{-200}) \approx 2.5\text{ A}$

Q. For the RL circuit shown in the Figure.

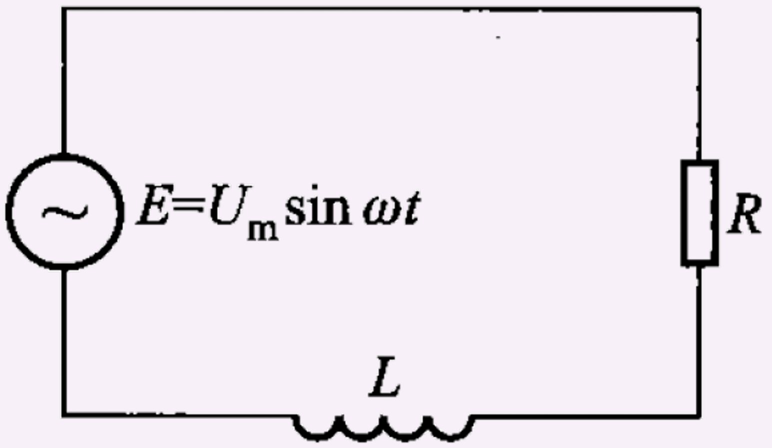
Find the variation law of the current $I(t)$ in the inductor L and explain its physical meaning.

(Assume $I = 0$ when $t = 0$).

Hint:

Replace E in the differential equation of current (Here, the value of E is changing by time), so this is the DE which describes how the inductor current changes over time during charging/discharging.

$$L\frac{\mathrm{d}I}{\mathrm{d}t} + RI = E \longrightarrow \boxed{L\frac{\mathrm{d}I}{\mathrm{d}t} + RI = U_m \sin \omega t}$$



the RL circuit

Solution:

$$L\frac{\mathrm{d}I}{\mathrm{d}t} + RI = U_m \sin\omega t$$

Divide it by L

$$\frac{\mathrm{d}I}{\mathrm{d}t} + \frac{R}{L}I = \frac{U_m}{L}\sin\omega t$$

Find integrating factor $\mu(t) = e^{\int \frac{R}{L}\,\mathrm{d}t} = e^{\frac{R}{L}t}$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(I(t)e^{\frac{R}{L}t}\right) = \frac{U_m}{L}e^{\frac{R}{L}t}\sin\omega t$$

Use integration table

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + c$$

Integration

$$I(t)e^{\frac{R}{L}t} = \frac{U_m e^{\frac{R}{L}t}(R\sin\omega t - \omega L\cos\omega t)}{R^2+(\omega L)^2} + c_1$$

General solution

$$I(t) = \frac{U_m(R\sin\omega t - \omega L\cos\omega t)}{R^2+(\omega L)^2} + c_1 e^{-\frac{R}{L}t}$$

Apply initial condition $I(0) = 0$

$$0 = \frac{U_m(0-\omega L\cdot 1)}{R^2+(\omega L)^2} + c_1$$

$$c_1 = \frac{U_m\omega L}{R^2+(\omega L)^2}$$

$$\boxed{I(t) = \frac{U_m(R\sin\omega t - \omega L\cos\omega t)}{R^2+(\omega L)^2} + \frac{U_m\omega L}{R^2+(\omega L)^2}e^{-\frac{R}{L}t}}$$

Q. Find the solution of the following differential equation, and discuss their interval of definitions

(1) $(x^2-1)y'-xy+1=0$

(2) $x(x^2-1)y'-(2x^2-1)y+x^3=0$

(3) $y'\sin x\cdot\cos x-y-\sin^3 x=0$

Solution:

Method: Change them to the standard form… $\dfrac{dy}{dx} + P(x)y = f(x)$,

(1) $\dfrac{dy}{dx} - \dfrac{x}{x^2-1}y = -\dfrac{1}{x^2-1}$

(2) $\dfrac{dy}{dx} - \dfrac{2x^2-1}{x(x^2-1)}y = -\dfrac{x^2}{x^2-1}$

(3) $\dfrac{dy}{dx} - \dfrac{1}{\sin x \cos x}y = \dfrac{\sin^2 x}{\cos x}$

Q. Find the solution of the following differential equations (Hint: use the exact DE method)

(1) $(x^2 + y)\mathrm{d}x + (x - 2y)\mathrm{d}y = 0$

(2) $(y - 3x^2)\mathrm{d}x - (4y - x)\mathrm{d}y = 0$

(3) $\left[\dfrac{y^2}{(x-y)^2} - \dfrac{1}{x}\right]\mathrm{d}x + \left[\dfrac{1}{y} - \dfrac{x^2}{(x-y)^2}\right]\mathrm{d}y = 0$

(4) $2(3xy^2 + 2x^3)\mathrm{d}x + 3(2x^2y + y^2)\mathrm{d}y = 0$

(5) $\left(\dfrac{1}{y}\sin\dfrac{x}{y} - \dfrac{y}{x^2}\cos\dfrac{y}{x} + 1\right)\mathrm{d}x + \left(\dfrac{1}{x}\cos\dfrac{y}{x} - \dfrac{x}{y^2}\sin\dfrac{x}{y} + \dfrac{1}{y^2}\right)\mathrm{d}y = 0$

Q. Find the solution of the following differential equations

(1) $\left(\cos x + \frac{1}{y}\right) \mathrm{d}x + \left(\frac{1}{y} - \frac{x}{y^2}\right) \mathrm{d}y = 0$ Hint: use the exact DE method

(2) $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y} + \sqrt{1 + \left(\frac{x}{y}\right)^2} \; (y > 0)$ Hint: define $u = \frac{x}{y}$ and remove x

Q. Find the solution of the following differential equations

(1) $2x\left(ye^{x^2}-1\right)\mathrm{d}x+e^{x^2}\mathrm{d}y=0$

(2) $\left(e^{x}+3y^2\right)\mathrm{d}x+2xy\mathrm{d}y=0$

(3) $2xy\mathrm{d}x+\left(x^2+1\right)\mathrm{d}y=0$

(4) $y\mathrm{d}x-x\mathrm{d}y=\left(x^2+y^2\right)\mathrm{d}x$

(5) $y\mathrm{d}x-\left(x+y^3\right)\mathrm{d}y=0$

(6) $(y-1-xy)\,\mathrm{d}x+x\mathrm{d}y=0$

(7) $\left(y-x^2\right)\mathrm{d}x-x\mathrm{d}y=0$

(8) $(x+2y)\,\mathrm{d}x+x\mathrm{d}y=0$

(9) $[x\cos(x+y)+\sin(x+y)]\,\mathrm{d}x+x\cos(x+y)\mathrm{d}y=0$

(10) $(y\cos x-x\sin x)\,\mathrm{d}x+(y\sin x+x\cos x)\,\mathrm{d}y=0$

(11) $x\,(4y\mathrm{d}x+2x\mathrm{d}y)+y^3\,(3y\mathrm{d}x+5x\mathrm{d}y)=0$

Q. Prove that the homogeneous differential equation

$$M(x,y)dx + N(x,y)dy = 0$$

has an integrating factor $\mu = \frac{1}{xM+yN}$

$(xM + yN \neq 0)$

Remark:

$M(x,y)dx + N(x,y)dy = 0$ is said to be a homogeneous DE if both M and N are homogeneous functions of the same degree.

M is a homogenous function $M(\lambda x, \lambda y) = \lambda^n M(x,y)$

N is a homogenous function $N(\lambda x, \lambda y) = \lambda^n N(x,y)$

Solution:

We need to show the equation after multiplying by the integrating factor is an exact differential equation.

$$\underbrace{\frac{M}{xM+yN}}_{P}dx+\underbrace{\frac{N}{xM+yN}}_{Q}dy=0$$

We have to show $\longrightarrow$ $\frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}$

$$\begin{aligned}\frac{\partial P}{\partial y}&=\frac{\partial}{\partial y}\left(\frac{M}{xM+yN}\right)\\&=\frac{\frac{\partial M}{\partial y}(xM+yN)-M\left(x\frac{\partial M}{\partial y}+N+y\frac{\partial N}{\partial y}\right)}{(xM+yN)^2}\\&=\frac{y\left(N\frac{\partial M}{\partial y}-M\frac{\partial N}{\partial y}\right)-MN}{(xM+yN)^2}.\end{aligned}$$

$$\begin{aligned}\frac{\partial Q}{\partial x}&=\frac{\partial}{\partial x}\left(\frac{N}{xM+yN}\right)\\&=\frac{\frac{\partial N}{\partial x}(xM+yN)-N\left(M+x\frac{\partial M}{\partial x}+y\frac{\partial N}{\partial x}\right)}{(xM+yN)^2}\\&=\frac{x\left(M\frac{\partial N}{\partial x}-N\frac{\partial M}{\partial x}\right)-MN}{(xM+yN)^2}.\end{aligned}$$

So, the proof will be done if we show $y\left(N\frac{\partial M}{\partial y}-M\frac{\partial N}{\partial y}\right)=x\left(M\frac{\partial N}{\partial x}-N\frac{\partial M}{\partial x}\right)$

Remark:

M is a homogenous function $\boxed{M(\lambda x, \lambda y) = \lambda^n M(x,y)}$

N is a homogenous function $\boxed{N(\lambda x, \lambda y) = \lambda^n N(x,y)}$

Find the differential of the above equations with respect λ,

Define $u = \lambda x,$ $v = \lambda y$ $\quad \frac{\partial M}{\partial u} \cdot \frac{du}{d\lambda} + \frac{\partial M}{\partial v} \cdot \frac{dv}{d\lambda} = M_u \cdot x + M_v \cdot y$

$$\longrightarrow \quad x\frac{\partial M}{\partial(\lambda x)} + y\frac{\partial M}{\partial(\lambda y)} = n\lambda^{n-1}M(x,y) \quad \longrightarrow \quad \boxed{xM_x(\lambda x, \lambda y) + yM_y(\lambda x, \lambda y) = n\lambda^{n-1}M(x,y)}$$

Similarly for N $\longrightarrow$ $\boxed{xN_x(\lambda x, \lambda y) + yN_y(\lambda x, \lambda y) = n\lambda^{n-1}N(x,y)}$

Now replace $\lambda = 1$

$$\Rightarrow \begin{cases} x\dfrac{\partial M}{\partial x} + y\dfrac{\partial M}{\partial y} = nM \\ x\dfrac{\partial N}{\partial x} + y\dfrac{\partial N}{\partial y} = nN \end{cases} \Rightarrow \begin{cases} xN\dfrac{\partial M}{\partial x} + yN\dfrac{\partial M}{\partial y} = nMN \\ xM\dfrac{\partial N}{\partial x} + yM\dfrac{\partial N}{\partial y} = nMN \end{cases}$$

By subtraction $$y\left(N\frac{\partial M}{\partial y} - M\frac{\partial N}{\partial y}\right) = x\left(M\frac{\partial N}{\partial x} - N\frac{\partial M}{\partial x}\right)$$

First-Order Implicit Differential Equations
&
Parametric Representation

Q. Find the solution of the following differential equation (Hint: let $\frac{dy}{dx} = p$)

$$\left(\frac{dy}{dx}\right)^3 + 2x\frac{dy}{dx} - y = 0,$$

Solution:

$$\left(\frac{dy}{dx}\right)^3 + 2x\frac{dy}{dx} - y = 0, \quad \xrightarrow{\frac{dy}{dx} = p} \quad y = p^3 + 2xp$$

Differentiate with respect to x

$$p = 3p^2\frac{dp}{dx} + 2p + 2x\frac{dp}{dx} \longrightarrow -p = \frac{dp}{dx}(3p^2 + 2x)$$

rarrangement

$$\frac{dx}{dp} = -\frac{3p^2+2x}{p}$$

By rarrangement $\quad \dfrac{dx}{dp} + \dfrac{2}{p}x = -3p$

Integrating factor $\quad \mu(p) = p^2 \longrightarrow \dfrac{d}{dp}(p^2x) = -3p^3 \longrightarrow p^2x = -\dfrac{3}{4}p^4 + c_1 \longrightarrow \boxed{x = -\dfrac{3}{4}p^2 + \dfrac{c_1}{p^2}}$

(now replace x in y) $\quad y = p^3 + 2xp$

Parametric solution $\rightarrow \begin{cases} x = -\dfrac{3}{4}p^2 + \dfrac{c_1}{p^2} \\ y = -\dfrac{1}{2}p^3 + \dfrac{2c_1}{p} \end{cases}$

Q. Find the solution of the following differential equation (Hint: let $\frac{dy}{dx} = p$)

$$y = \left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + \frac{x^2}{2}$$

Solution:

$$y = \left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + \frac{x^2}{2} \xrightarrow{\frac{dy}{dx} = p} y = p^2 - xp + \frac{x^2}{2}$$

Differentiate with respect to x

$$\underbrace{\cancel{\frac{dy}{dx}}}_{p} = 2p\frac{dp}{dx} - p - x\frac{dp}{dx} + x \longrightarrow p + p - x = (2p - x)\frac{dp}{dx}$$

Rearrangement

$$(2p - x)\left(1 - \frac{dp}{dx}\right) = 0$$

There are two possible senarios: $2p = x$ $\quad \frac{dp}{dx} = 1$

$$y = \left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + \frac{x^2}{2}$$

Case 1. $2p - x = 0 \longrightarrow \dfrac{dy}{dx} = \dfrac{x}{2} \longrightarrow \boxed{y = \dfrac{x^2}{4} + c_1}$

Case 2. $\dfrac{dp}{dx} = 1 \longrightarrow p = x + c_2 \longrightarrow \dfrac{dy}{dx} = x + c_2$

$$\longrightarrow y = \frac{x^2}{2} + c_2 x + c_3$$

The equation is a first-order DE, so there must be only one free parameter in the general solution;
so we replace y in the DE to see how we can remove either c_2 or c_3

Replacement $\dfrac{x^2}{2} + c_2 x + c_3 = (x + c_2)^2 - x\,(x + c_2) + \dfrac{x^2}{2}$

$$\frac{x^2}{2} + c_2 x + c_3 = \frac{x^2}{2} + c_2 x + c_2^2 \longrightarrow c_3 = c_2^2$$

$$\boxed{y = \frac{x^2}{2} + c_2 x + c_2^2}$$

Q. Find the solution of the following differential equation (Hint: let $\frac{dy}{dx} = p = tx$)

$$x^3 + (y')^3 - 3xy' = 0,$$

Solution:

$$x^3 + (y')^3 - 3xy' = 0,$$

$$\xrightarrow{\frac{dy}{dx} = p} \quad x^3 + p^3 - 3xp = 0$$

Again substitute $p = tx$

$$x^3 + (tx)^3 - 3x(tx) = 0$$

$$x^2(x + t^3x - 3t) = 0 \longrightarrow x = \frac{3t}{1+t^3}$$

Note that if $x = 0$, then $p = tx = 0$, so, $y' = 0 \rightarrow y = c_1$ (also a solution)

$$\xrightarrow{\text{Also,}} \quad p = tx = \frac{3t^2}{1+t^3} \xrightarrow{dy = pdx} dy = \frac{3t^2}{1+t^3} \cdot \overbrace{\frac{3(1-2t^3)}{(1+t^3)^2}dt}^{dx} = \frac{9t^2(1-2t^3)}{(1+t^3)^3}dt$$

$$\xrightarrow{\text{Integration}} \quad y = \frac{6}{1+t^3} - \frac{9}{2(1+t^3)^2} + c$$

So, the general solution in the parametric solution is:

$$\boxed{\begin{aligned} x &= \frac{3t}{1+t^3} \\ y &= \frac{6}{1+t^3} - \frac{9}{2(1+t^3)^2} + c \end{aligned}}$$

Q. Find the solution of the following differential equation (Hint: let $2 - y' = ty$)

$$y^2(1 - y') = (2 - y')^2$$

Solution:

$$y^2(1-y') = (2-y')^2$$

substitute $2 - y' = ty$ $\xrightarrow{y' = 2 - ty}$

$$y^2(ty-1) = t^2y^2 \longrightarrow y = \frac{1}{t} + t$$

& what is x ?

$$\xrightarrow{y' = \frac{dy}{dx}} dx = \frac{dy}{y'} \longrightarrow dx = \frac{\left(1 - \frac{1}{t^2}\right)dt}{2 - ty} = \frac{\left(1 - \frac{1}{t^2}\right)dt}{1 - t^2} = -\frac{1}{t^2}$$

(general solution, parametric form) $\longrightarrow \begin{cases} x = \dfrac{1}{t} + c_1 \\ y = \dfrac{1}{t} + t \end{cases}$

Q. Find the solution of the following differential equation

$$xy'^3 = 1 + y'$$

Solution:

$$xy'^3 = 1 + y' \xrightarrow{\frac{dy}{dx} = p} xp^3 = 1 + p \longrightarrow x = \frac{1+p}{p^3} = \frac{1}{p^3} + \frac{1}{p^2} \quad (p \neq 0) \qquad \& \; y\,?$$

$$\xrightarrow{p = \frac{dy}{dx}} dy = pdx \xrightarrow{dx = \left(-\frac{3}{p^4} - \frac{2}{p^3}\right) dp} dy = p\left(-\frac{3}{p^4} - \frac{2}{p^3}\right) dp = \left(-\frac{3}{p^3} - \frac{2}{p^2}\right) dp$$

$$\longrightarrow y = \frac{3}{2p^2} + \frac{2}{p} + c_1$$

(general solution, parametric form) $\longrightarrow$

$$\begin{cases} x = \dfrac{1+p}{p^3} = \dfrac{1}{p^3} + \dfrac{1}{p^2} \\ y = \dfrac{3}{2p^2} + \dfrac{2}{p} + c_1 \end{cases}$$

Q. Find the solution of the following differential equation

$$y'^3 - x^3(1 - y') = 0$$

Solution :

$$y'^3 - x^3(1-y') = 0, \xrightarrow{\frac{dy}{dx}=p} p^3 - x^3(1-p) = 0 \longrightarrow x = \frac{p}{(1-p)^{1/3}} \quad (p \neq 1) \quad \& \; y\,?$$

$$\xrightarrow{p=\frac{dy}{dx}} dy = p\,dx \xrightarrow{dx = \frac{3-2p}{3(1-p)^{4/3}}dp} dy = p\cdot\frac{3-2p}{3(1-p)^{4/3}}dp = \frac{p(3-2p)}{3(1-p)^{4/3}}dp$$

$$\longrightarrow y = -(1-p)^{-1/3} + \frac{1}{2}(1-p)^{2/3} - \frac{2}{5}(1-p)^{5/3} + C$$

(general solution, parametric form) $\longrightarrow \begin{cases} x = \dfrac{p}{(1-p)^{1/3}} \\ y = -(1-p)^{-1/3} + \dfrac{1}{2}(1-p)^{2/3} - \dfrac{2}{5}(1-p)^{5/3} + C \end{cases}$

Q. Find the solution of the following differential equation

$$y = y'^2 e^{y'}$$

Solution :

$$y = {y'}^2 e^{y'} \xrightarrow{\frac{dy}{dx} = p} y = p^2 e^p \qquad x\ ?$$

$$\xrightarrow{p = \frac{dy}{dx}} dy = p\,dx \xrightarrow{dy = p(2+p)e^p\,dp} \cancel{p}\,dx = \cancel{p}(2+p)e^p\,dp$$

$$\longrightarrow x = (1+p)e^p + c_1$$

(general solution, parametric form) $\longrightarrow \begin{cases} x = (1+p)e^p + c_1 \\ y = p^2 e^p \end{cases}$

Q. Find the solution of the following differential equation

$$y(1 + y'^2) = 2a, \qquad a \text{ is a given constant}$$

Solution :

$$y(1+y'^2)=2a, \xrightarrow{\frac{dy}{dx}=p} y(1+p^2)=2a \longrightarrow y=\frac{2a}{1+p^2} \qquad \& \ x\,?$$

$$\xrightarrow{p=\frac{dy}{dx}} dy=pdx \xrightarrow{dy=\frac{-4ap}{(1+p^2)^2}dp} dx=\frac{-4a}{(1+p^2)^2}dp \longrightarrow x=-4a\int\frac{dp}{(1+p^2)^2}+c$$

$$=-4a\cdot\frac{1}{2}\left(\frac{p}{1+p^2}+\arctan p\right)+c$$

$$=-2a\left(\frac{p}{1+p^2}+\arctan p\right)+c$$

(general solution, parametric form)
$$\begin{cases} x=-2a\left(\dfrac{p}{1+p^2}+\arctan p\right)+c \\ y=\dfrac{2a}{1+p^2} \end{cases}$$

Integration:

Let $p = \tan\theta,\ dp = \sec^2\theta\, d\theta,\ 1 + p^2 = \sec^2\theta$

$$\int \frac{1}{(1+p^2)^2} dp = \int \cos^2\theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta = \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta + c_1$$

$$\theta = \arctan p,\ \sin 2\theta = \frac{2p}{1+p^2}$$

$$\int \frac{1}{(1+p^2)^2}\, dp = \frac{1}{2}\arctan p + \frac{p}{2(1+p^2)} + c_1$$

Solution (5):

$$x^2 + y'^2 = 1 \xrightarrow{\frac{dy}{dx} = p} x^2 + p^2 = 1 \longrightarrow x = \pm\sqrt{1 - p^2} \quad \& \ y\,?$$

$$\xrightarrow{p = \frac{dy}{dx}} dy = p\,dx \xrightarrow{dx = \mp\frac{p}{\sqrt{1-p^2}}\,dp} dy = p \cdot \left(\mp\frac{p}{\sqrt{1-p^2}}\,dp\right) = \mp\frac{p^2}{\sqrt{1-p^2}}\,dp$$

Integration $\quad \frac{p^2}{\sqrt{1-p^2}} = -\sqrt{1-p^2} + \frac{1}{\sqrt{1-p^2}}$

$$\longrightarrow y = \pm\tfrac{1}{2}\left(p\sqrt{1-p^2} - \arcsin p\right) + c_1$$

(general solution, parametric form) $\longrightarrow$

$$\begin{cases} x = \pm\sqrt{1-p^2} \\ y = \pm\dfrac{1}{2}\left(p\sqrt{1-p^2} - \arcsin p\right) + c_1 \end{cases}$$

More Exercises:

Find the solution of the given DEs

(1) $y \sin x + \dfrac{dy}{dx} \cos x = 1$

(2) $y\, dx - x\, dy = x^2 y\, dy$

(3) $\dfrac{dy}{dx} = 4e^{-y} \sin x - 1$

(4) $\dfrac{dy}{dx} = \dfrac{y}{x - \sqrt{xy}}$

(5) $\left(xye^{\frac{x}{y}} + y^2\right) dx - x^2 e^{\frac{x}{y}} dy = 0$

(6) $(xy + 1)y\, dx - x\, dy = 0$

(7) $(2x + 2y - 1)dx + (x + y - 2)dy = 0$

(8) $\dfrac{dy}{dx} = \dfrac{y}{x} + \dfrac{y^2}{x^3}$

(9) $\dfrac{dy}{dx} = 3y + x - 2$

(10) $x\dfrac{dy}{dx} = 1 + \left(\dfrac{dy}{dx}\right)^2$

(11) $\dfrac{dy}{dx} = \dfrac{x - y + 1}{x + y^2 + 3}$

(12) $e^{-y}\left(\dfrac{dy}{dx} + 1\right) = xe^x$

(13) $(x^2 + y^2)dx - 2xydy = 0$

(14) $\dfrac{dy}{dx} = x + y + 1$

(15) $\dfrac{dy}{dx} = e^{\frac{y}{x}} + \dfrac{y}{x}$

(16) $(x + 1)\dfrac{dy}{dx} + 1 = 2e^{-y}$

(17) $(x-y^2)dx+y(1+x)dy=0$

(18) $4x^2y^2dx+2(x^3y-1)dy=0$

(19) $x\left(\frac{dy}{dx}\right)^2-2y\left(\frac{dy}{dx}\right)+4x=0$

(20) $y^2\left[1-\left(\frac{dy}{dx}\right)^2\right]=1$

(21) $\left(1+\mathrm{e}^{\frac{x}{y}}\right)dx+\mathrm{e}^{\frac{x}{y}}\left(1-\frac{x}{y}\right)dy=0$

(22) $\frac{2x}{y^3}dx+\frac{y^2-3x^2}{y^4}dy=0$

(23) $ydx-(1+x+y^2)dy=0$

(24) $\left[y-x(x^2+y^2)\right]dx-xdy=0$

(25) $\frac{dy}{dx}+\mathrm{e}^{\frac{dy}{dx}}-x=0$

(26) $\left(2xy+x^2y+\frac{y^3}{3}\right)dx+(x^2+y^2)dy=0$

(27) $\frac{dy}{dx}=\frac{2x+3y+4}{4x+6y+5}$

(28) $x\frac{dy}{dx}-y=2x^2y(y^2-x^2)$

(29) $\frac{dy}{dx}+\frac{y}{x}=\mathrm{e}^{xy}$

(30) $\frac{dy}{dx}=\frac{4x^3-2xy^3+2x}{3x^2y^2-6y^5+3y^2}$

Q. Find the solution of the following differential equation

$$y^2(xdx + ydy) + x(ydx - xdy) = 0$$

(Hint: let $x = \rho\cos\theta$, $y = \rho\sin\theta$)

(Hint: let $\frac{x}{y} = u$)

Solution :

$$x=\rho\cos\theta,\ y=\rho\sin\theta \longrightarrow \rho^2=x^2+y^2 \longrightarrow \rho d\rho = xdx+ydy$$

$$y^2\underbrace{\cancel{(xdx+ydy)}}_{\rho d\rho}+x\underbrace{\cancel{(ydx-xdy)}}_{-\rho^2 d\theta}=0 \xrightarrow[\begin{cases}x=\rho\cos\theta\\ y=\rho\sin\theta\end{cases}]{\text{Replace}} \rho^3\sin^2\theta\,d\rho-\rho^3\cos\theta\,d\theta=0$$

$$\longrightarrow \sin^2\theta\,d\rho=\cos\theta\,d\theta$$

$$\longrightarrow d\rho=\frac{\cos\theta}{\sin^2\theta}d\theta$$

$$\longrightarrow \boxed{\rho=-\frac{1}{\sin\theta}+c_1}$$

$$\begin{aligned}
ydx-xdy &= \rho\sin\theta\cdot d(\rho\cos\theta)-\rho\cos\theta\cdot d(\rho\sin\theta)\\
&=\rho\sin\theta(\cos\theta d\rho-\rho\sin\theta d\theta)-\rho\cos\theta(\sin\theta d\rho+\rho\cos\theta d\theta)\\
&=\rho\cos\theta\sin\theta d\rho-\rho^2\sin^2\theta d\theta-\rho\cos\theta\sin\theta d\rho-\rho^2\cos^2\theta d\theta\\
&=-\rho^2(\sin^2\theta+\cos^2\theta)d\theta\\
&=-\rho^2 d\theta
\end{aligned}$$

$$\rho^2 = x^2 + y^2 \quad \& \quad \begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} \longrightarrow \rho = \sqrt{x^2+y^2},\ \sin\theta = \tfrac{y}{\rho} = \tfrac{y}{\sqrt{x^2+y^2}}$$

Replace in $\rho = \dfrac{1}{\sin\theta} + c_1$

$$\sqrt{x^2+y^2} + \frac{\sqrt{x^2+y^2}}{y} = c_1$$

$$(y+1)\sqrt{x^2+y^2} = c_1 y$$

(general solution, imilicit form)

Q. Find the solution of the following differential equation (Hint: let $u = x + y, v = xy$)

$$\frac{dy}{dx} + \frac{1 + xy^3}{1 + x^3 y} = 0$$

Picard Iteration Formula

For IVP: $$y' = f(x, y), \qquad y(x_0) = y_0$$

The solution approximately can be obtaind by

$$y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))dt\,, \qquad n = 0,1,2, \dots$$

As $n \to \infty, \;\; y_n\,(x) \to y$ (solution)

It is namely the Picard Iteration Formula

Where is this idea coming from?

If we integrate both sides of $y' = f(x, y),\; y(x_0) = y_0$, so we have

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))dt$$

Let, $y_0\,(x) = y_0$, then

$$y_1(x) = y_0 + \int_{x_0}^{x} f(t, y_0(t))dt$$

$$y_2(x) = y_0 + \int_{x_0}^{x} f(t, y_1(t))dt$$

$$y_3(x) = y_0 + \int_{x_0}^{x} f(t, y_2(t))dt$$

…

When and Why does it work?

Continuity: $f(x, y)$ must be continuous on a rectangular region around (x_0, y_0).

This guarantees the integral in the iteration is well-defined and exists

Lipschitz Condition: $|f(x, y_2) - f(x, y_1)| < L|y_2 - y_1|$ for some constant L (Lipschitz constant).

It prevents the iterative sequence $\{y_n(x)\}$ from diverging

Example: for $y' = x + y,$ and $y(0) = 0$, apply the picard iteration formula

$$y_0(x) = 0$$

$$y_1(x) = 0 + \int_0^x (t+0)dt = \frac{x^2}{2}$$

$$y_2(x) = 0 + \int_0^x \left(t + \frac{t^2}{2}\right) dt = \frac{x^2}{2} + \frac{x^3}{6}$$

$$y_3(x) = 0 + \int_0^x \left(t + \frac{t^2}{2} + \frac{t^3}{6}\right) dt = \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$$

$$y(x) = e^x - x - 1$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

Example: Find the third approximate solution of the equation $y' = x + y^2$, passing through the point (0,0)

$$y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))dt\,, \qquad y_0\,(x_0) = y_0$$

Step 1. Initial Iteration ($n = 0$) $\quad x_0 = 0, y_0 = 0 \;\rightarrow\; y_0\,(x) = 0$

Step 2. First Iteration ($n = 1$)

$$y_1(x) = 0 + \int_0^x f(t, y_0(t))\,dt$$

$$= \int_0^x (t + 0^2)\,dt$$

$$= \int_0^x t\,dt$$

$$= \frac{x^2}{2}$$

$y' = f(x, y)$

Step 3. Second Iteration ($n = 2$)

$$
\begin{aligned}
y_2(x) &= 0 + \int_0^x f(t, y_1(t))\, dt \\
&= \int_0^x \left(t + \left(\frac{t^2}{2} \right)^2 \right) dt \\
&= \int_0^x \left(t + \frac{t^4}{4} \right) dt \\
&= \frac{x^2}{2} + \frac{x^5}{20}
\end{aligned}
$$

Step 4. Third Iteration ($n = 3$)

$$
\begin{aligned}
y_3(x) &= 0 + \int_0^x f(t, y_2(t))\, dt \\
&= \int_0^x \left(t + \left(\frac{t^2}{2} + \frac{t^5}{20} \right)^2 \right) dt \\
&= \int_0^x \left(t + \frac{t^4}{4} + \frac{t^7}{20} + \frac{t^{10}}{400} \right) dt \\
&= \frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160} + \frac{x^{11}}{4400}
\end{aligned}
$$

The third approximate solution of $y' = x + y^2$, passing through the point (0,0) is

$$y_3(x) = \frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160} + \frac{x^{11}}{4400}$$

Example: Find the second approximate solution of the equation $y' = x - y^2$, passing through the point (1,0)

$$y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))dt\,, \qquad y_0\,(x_0) = y_0$$

Step 1. Initial Iteration ($n = 0$) $\quad x_0 = 1, y_0 = 0 \;\rightarrow\; y_0\,(x) = 0$

$y' = f(x, y)$

Step 2. First Iteration ($n = 1$)

$$\begin{aligned} y_1(x) &= 0 + \int_1^x f(t, y_0(t))\,dt \\ &= \int_1^x (t - 0^2)\,dt \\ &= \int_1^x t\,dt \\ &= \frac{x^2}{2} - \frac{1^2}{2} \\ &= \frac{x^2}{2} - \frac{1}{2} \end{aligned}$$

Step 3. Second Iteration ($n = 2$)

$$
\begin{aligned}
y_2(x) &= 0 + \int_1^x f(t, y_1(t))dt \\
&= \int_1^x \left(t - \left(\frac{t^2}{2} - \frac{1}{2}\right)^2\right) dt \\
&= \int_1^x \left(t - \left(\frac{t^4}{4} - \frac{t^2}{2} + \frac{1}{4}\right)\right) dt \\
&= \int_1^x \left(-\frac{t^4}{4} + \frac{t^2}{2} + t - \frac{1}{4}\right) dt \\
&= \left[-\frac{t^5}{20} + \frac{t^3}{6} + \frac{t^2}{2} - \frac{t}{4}\right]_1^x \\
&= \left(-\frac{x^5}{20} + \frac{x^3}{6} + \frac{x^2}{2} - \frac{x}{4}\right) - \left(-\frac{1}{20} + \frac{1}{6} + \frac{1}{2} - \frac{1}{4}\right) \\
&= -\frac{x^5}{20} + \frac{x^3}{6} + \frac{x^2}{2} - \frac{x}{4} + \frac{1}{20} - \frac{1}{6} - \frac{1}{2} + \frac{1}{4} \\
&= -\frac{x^5}{20} + \frac{x^3}{6} + \frac{x^2}{2} - \frac{x}{4} - \frac{7}{30}
\end{aligned}
$$

The second approximate solution of $y' = x - y^2$, passing through the point (1,0) is

$$y_2(x) = -\frac{x^5}{20} + \frac{x^3}{6} + \frac{x^2}{2} - \frac{x}{4} - \frac{7}{30}$$

Example: Find the second approximate solution of the equation $y' = x^2 - y^2$, passing through the point (-1,0)

$$\boxed{y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))dt, \qquad y_0(x_0) = y_0}$$

Step 1. Initial Iteration ($n = 0$) $x_0 = -1, y_0 = 0 \rightarrow y_0 = 0$

Step 2. First Iteration ($n = 1$)

$$\begin{aligned} y_1(x) &= 0 + \int_{-1}^{x} (t^2 - 0^2)dt \\ &= \int_{-1}^{x} t^2 dt \\ &= \frac{(x+1)^3}{3} - \frac{1}{3} \end{aligned}$$

$y' = f(x, y)$

Step 3. Second Iteration ($n = 2$)

$$y_2(x) = 0 + \int_{-1}^{x} \left(t^2 - \left(\frac{(t+1)^3}{3} - \frac{1}{3} \right)^2 \right) dt$$

$$= \int_{-1}^{x} \left(t^2 - \frac{(t+1)^6}{9} + \frac{2(t+1)^3}{9} - \frac{1}{9} \right) dt$$

$$= \frac{(x+1)^3}{3} - \frac{1}{9} \cdot \frac{(x+1)^7}{7} + \frac{2}{9} \cdot \frac{(x+1)^4}{4} - \frac{1}{9}(x+1) \Big|_{-1}^{x}$$

$$= \frac{(x+1)^3}{3} - \frac{(x+1)^7}{63} + \frac{(x+1)^4}{18} - \frac{x+1}{9}$$

The second approximate solution of $y' = x^2 - y^2$, passing through the point (-1,0) is

$$y_2(x) = \frac{(x+1)^3}{3} - \frac{(x+1)^7}{63} + \frac{(x+1)^4}{18} - \frac{x+1}{9}$$